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AIDA
Collaboration | Innovation | Excellence

DISTRIBUTION UTILITY MEET
DUM 2026

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TATA POWER

Chandigarh Power Distribution Limited (CPDL)

TATA POWER-DDL
Towards a greener tomorrow

BSES
BSES Rajdhani Power Limited

NPCL
NOIDA POWER COMPANY LIMITED

BSES
BSES Yamuna Power Limited

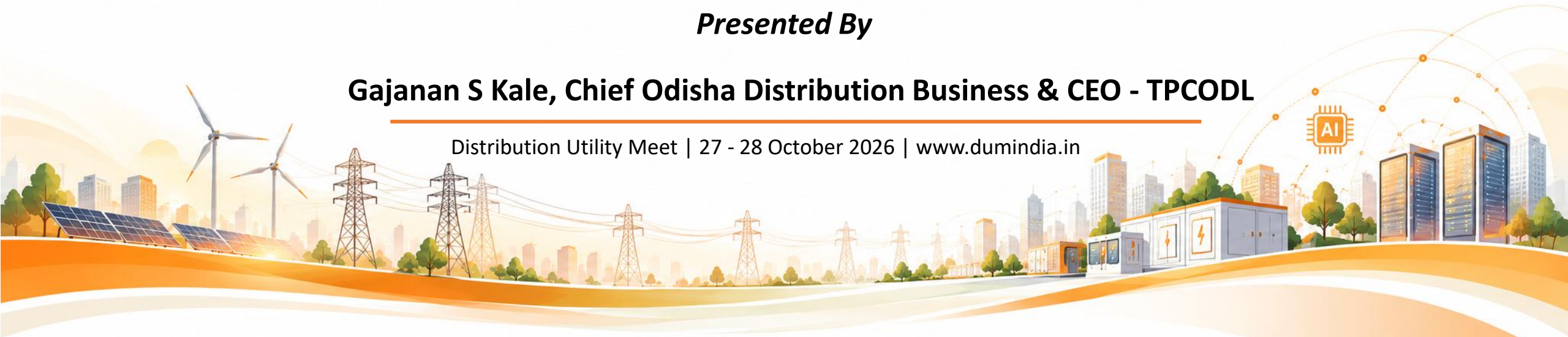
DUM 2026 CONFERENCE SESSION

SESSION 5: CLIMATE RESILIENCE AND SECURING THE PHYSICAL ASSETS OF DISCOMs

Presented By

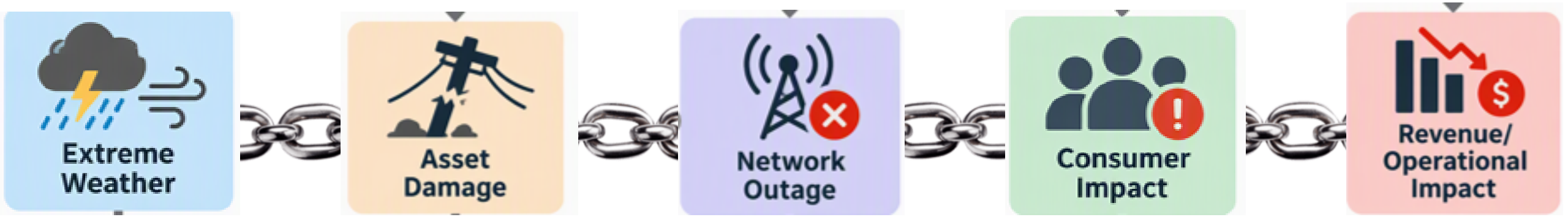
Gajanan S Kale, Chief Odisha Distribution Business & CEO - TPCODL

Distribution Utility Meet | 27 - 28 October 2026 | www.dumindia.in



Why Climate Resilience Matters for a DISCOM

Climate events & linked Operational challenge



Key hazards for Odisha

- Cyclones & high-velocity winds
- Extreme rainfall & flooding
- Lightning & thunderstorms
- Coastal inundation / saline environment
- Extreme heat
- Vegetation-related failures

Distribution Infrastructure

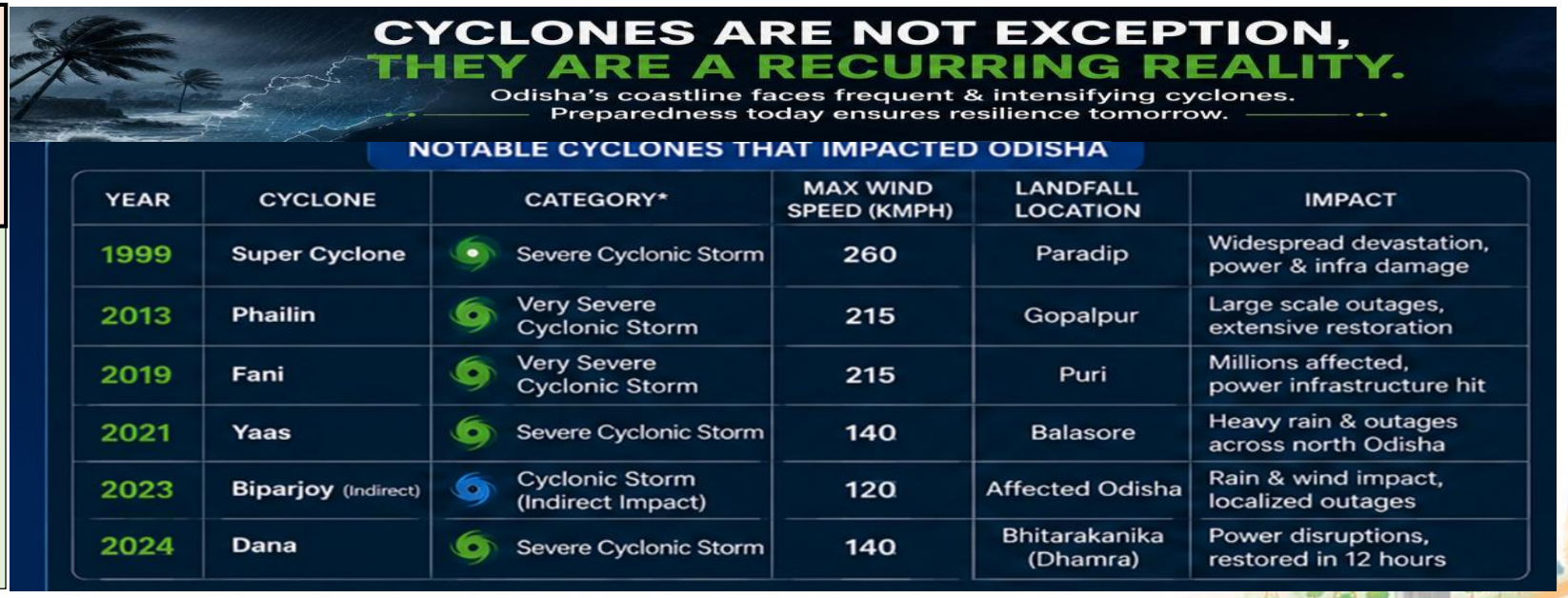
The first and most exposed layer of the power system during extreme weather

Odisha's Climate Risk Landscape

Climate Risk	Vulnerable Assets	Typical Impact
Cyclone / High Wind	Poles, conductors, DTs, substations	Pole failure, conductor snapping
Flooding	PSS, DTs, RMUs, cables	Water ingress, equipment damage
Lightning	Transformers, lines, protection equipment	Flashover/equipment failure
Coastal Exposure	Poles, structures, equipment	Corrosion / deterioration
Extreme Heat	Transformers, Cables, Switchgear	Thermal stress
Wind with Rampant Vegetation	Overhead lines	High Damage Faults & interruptions

Climate risk is
- Asset-specific
- Location-specific

Need proactive enablement by:-
- Technological intervention for vulnerability identification
- Resilience planning targeting impacted geographies



YEAR	CYCLONE	CATEGORY*	MAX WIND SPEED (KMPH)	LANDFALL LOCATION	IMPACT
1999	Super Cyclone	Severe Cyclonic Storm	260	Paradip	Widespread devastation, power & infra damage
2013	Phailin	Very Severe Cyclonic Storm	215	Gopalpur	Large scale outages, extensive restoration
2019	Fani	Very Severe Cyclonic Storm	215	Puri	Millions affected, power infrastructure hit
2021	Yaas	Severe Cyclonic Storm	140	Balasore	Heavy rain & outages across north Odisha
2023	Biparjoy (Indirect)	Cyclonic Storm (Indirect Impact)	120	Affected Odisha	Rain & wind impact, localized outages
2024	Dana	Severe Cyclonic Storm	140	Bhitarakanika (Dhamra)	Power disruptions, restored in 12 hours

From “Repair After Failure” to “Resilience by Design”

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Traditional Approach



Resilient Approach



Basic Principle for adopting RESILIENCE

Prevent

- Strengthen vulnerable assets

Prepare

- Identify risk zones and pre-position resources

Respond

- Enable remote monitoring, switching and coordinated restoration

Recover & Improve

- Analyze damage and redesign vulnerable infrastructure

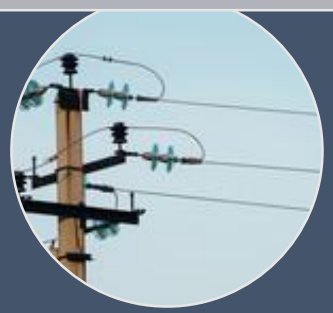
“Resilience is not only about stronger assets —
it is about reducing the time between hazard occurrence and restoration.”

Engineering Design Interventions - “Building Stronger Assets”



Poles & Structures

- Wind-resistant pole/structure design
- Improved foundation in vulnerable areas
- Strengthening of critical structures
- Reduction of vulnerable sections



Conductors & Lines

- Improved conductor/insulator selection
- Strengthening of critical crossings
- Vegetation management
- Strategic undergrounding



Transformers

- Improved anchoring
- Elevated installation in flood-prone areas
- Protection against water ingress
- Critical DT identification and redundancy



Substations / PSS

- Flood protection / elevation
- Water drainage
- Equipment protection
- Emergency backup arrangements



RMUs / Switchgear

- Weather-resistant enclosures
- Flood-prone asset elevation
- Remote monitoring / operation

Asset Criticality + Climate Exposure + Consumer Impact Criticality → Resilience Priority → Risk Based Investment Approach

Resilience Improvement Journey in Pole Design



WPB Pole
Design Wind Speed-
200 KMPH
FY 21-22



H Pole
Design Wind Speed-
250KMPH
FY 22-23



RLP Pole
Design Wind Speed-
300KMPH
FY 23-24



FRP Pole
Design Wind speed-
300KMPH
FY 25-26

Mitigating Low Voltage Issues in Heat wave through LT BESS

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LT TAIL END VOLTAGE BEFORE BESS INSTALLATION

145.4 V

Average LT voltage

LT TAIL END VOLTAGE AFTER BESS INSTALLATION

208.6 V

Average LT voltage

VOLTAGE IMPROVEMENT

+63.2 V

≈43% increase vs. baseline

THE CHALLENGE

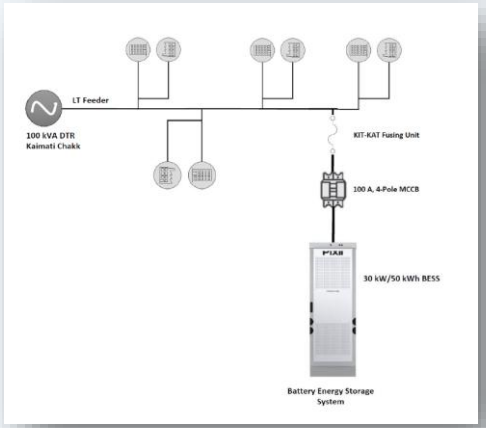
During April–May heatwaves, the average voltage at the tail end consumer in certain pockets drops to **145.4 V**, significantly below the standard **230 V**, affecting power supply quality.

THE INTERVENTION

A pole-mounted BESS commissioned to support the LT feeder and improve tail-end voltage during this summer peak demand.

30 kW / 50 kWh Pole-Mounted BESS at Dhenkanal Site

Whenever the LT voltage dips, the BESS provides voltage support to maintain stable voltage levels and improve power quality at the feeder tail-end.





Key Benefits of Pole-Mounted LT BESS

- Higher Voltage at Tail End
- Supports Peak Demand During Heat Waves
- Energy Storage for Local Support
- May Defer Network Augmentation
- Targeted and Scalable Deployment

This enables power quality improvement through voltage stabilization at the feeder tail end and supply reliability during peak demand

Technology Intervention: GIS-Based Climate Risk Mapping 1/2

WEATHER INTELLIGENCE & ASSET DATABASE INTEGRATED GEOSPATIAL SOLUTION

Unified Data. Unified Decisions. Unified Success.

INTEGRATED DATA LAYERS

IMD WEATHER FORECASTS
Real-time cyclone tracks, wind speed & impact zones



SURVEY OF INDIA VILLAGE BOUNDARIES
Accurate village/town boundaries & administrative maps



ENTERPRISE GIS – TPCODL ASSETS
Network topology – feeders, lines, DTs, substations & consumers



OPERATIONS DATA
Outages, crew status, resource availability, inventory & logistics



GEO-FENCED PERMIT TO WORK (LAKSHMAN REKHA)
Real-time location verification of crew against authorised work zone



INTELLIGENCE & DECISION ENGINE

GEO-SPATIAL DECISION ENGINE
Analytics | AI/ML | Rules

OPERATIONS DASHBOARD @ TPCODL CONTROL CENTRE



ACTION & FIELD EXECUTION

IMPACT ASSESSMENT
Predicts areas, assets & consumers likely to be impacted

PRIORITIZATION
Prioritizes feeders, villages & critical installations

RESOURCE OPTIMIZATION
Optimal deployment of manpower, materials & machinery

CRITICAL INFRA MAPPING
Hospitals, admin offices, water supply, oxygen plants & other essential services

SAFETY ASSURANCE
Geo-fenced PTW ensures right person at right location for safe execution

EARLY ALERTS
Timely alerts to teams & stakeholders

PREPAREDNESS
Pre-position teams, materials & equipment

COORDINATION
Seamless coordination with District Admin, NDRF, Fire Brigade & other agencies

RAPID RESTORATION
Faster fault rectification & power restoration

SAFE OPERATIONS
Digital verification reduces risk & ensures compliance

RESILIENT NETWORK. SAFE WORKFORCE. HAPPY COMMUNITIES.

THE POWER OF INTEGRATION

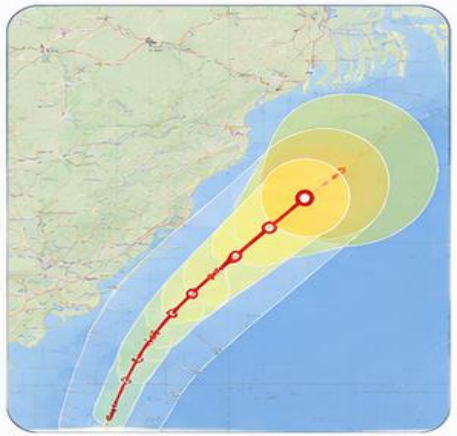
- Single Source of Truth
- Data-Driven Decisions
- Reduced Downtime & Faster Restoration
- Optimized Resources & Lower Cost
- Enhanced Safety & Sustainability

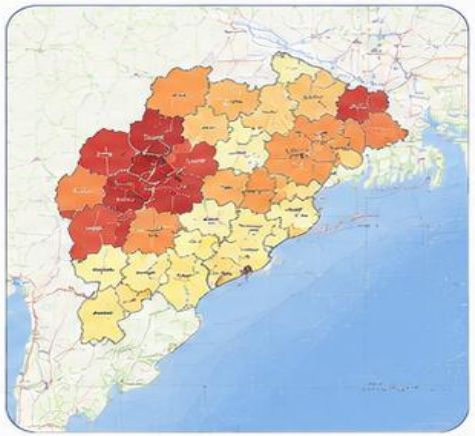
Technology Intervention: GIS-Based Climate Risk Mapping 2/2

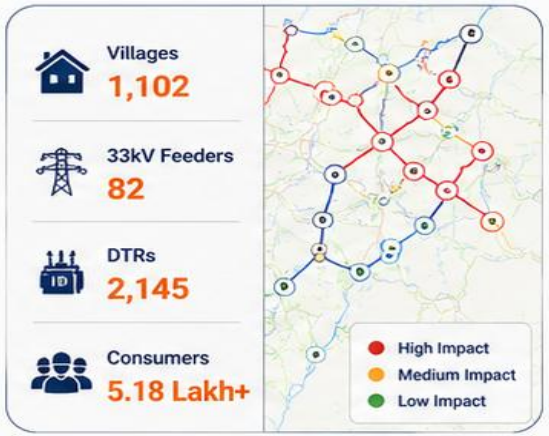
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01 WEATHER FORECAST → **02 GIS OVERLAY & IMPACT ANALYSIS** → **03 AFFECTED VILLAGES & FEEDERS** → **04 RESOURCE MOBILIZATION** → **05 FASTER RESTORATION**











IMD real-time forecasts and cyclone trajectory with wind speed zones

Overlay of IMD data with Survey of India maps & TPCODL network (GIS)

Granular impact assessment at village, feeder & asset level with severity classification

Proactive deployment of men, material & machinery to the right location

Faster, safer restoration ensuring reliable power to communities

Predictive Planning | **Data-driven Decisions** | **Optimized Resources** | **Safer Operations** | **Happier Communities**

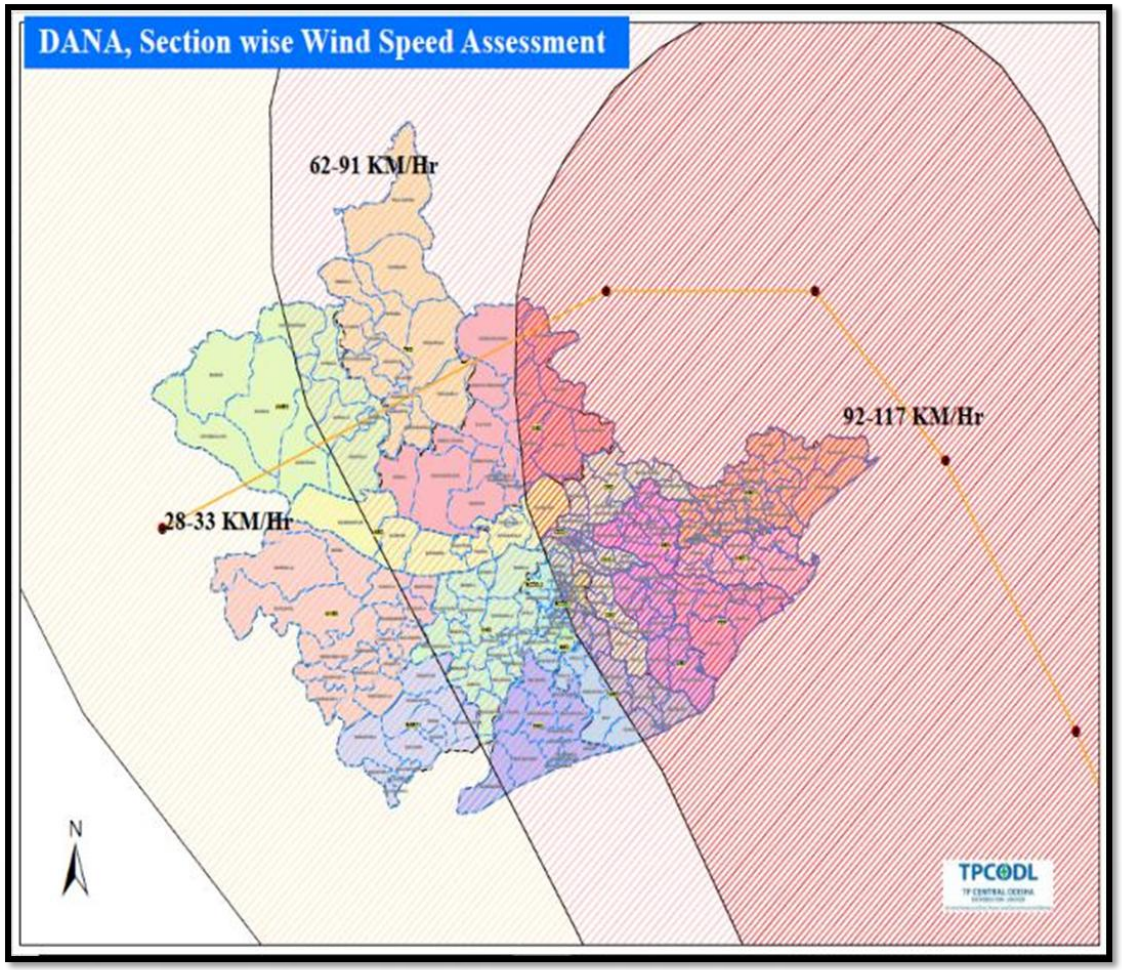
ONE PLATFORM. ONE VIEW. ONE PURPOSE – RESILIENCE & SAFETY FOR ALL.

Weather Forecast + Cyclone Track / Wind & Rainfall Forecast + Utility Network + Critical Assets & Consumers

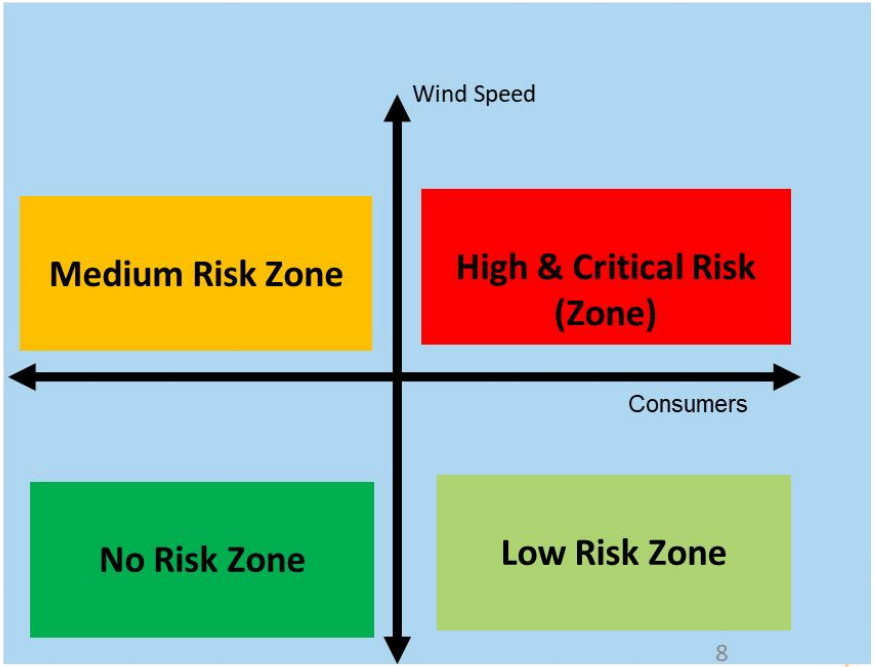
⟶ Climate Vulnerability Map ⟶ Priority Restorative Actions

Case Study – Dana Cyclone

Location based categorization of network vulnerability assessment

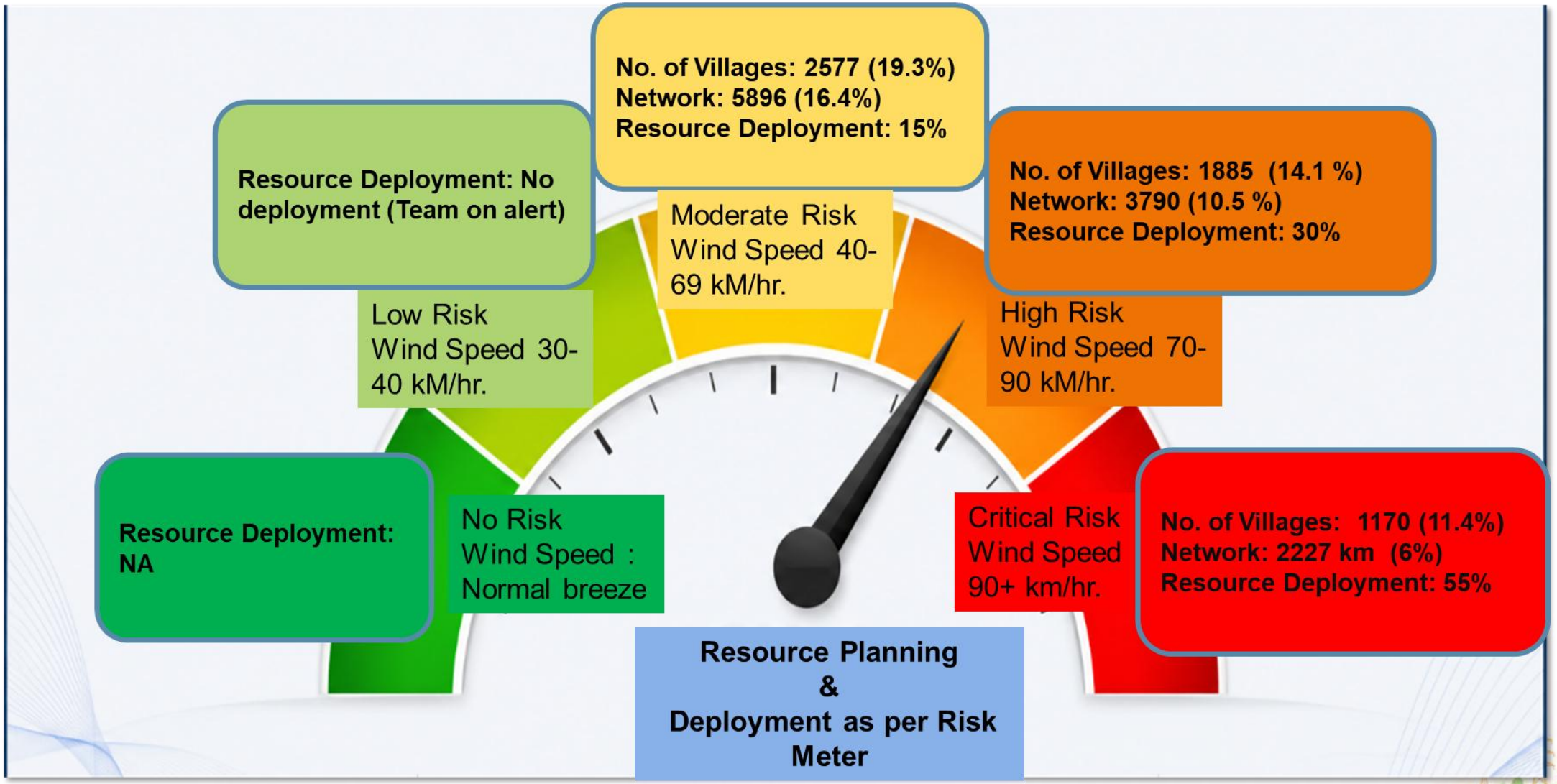


Categorization of villages & distribution network most vulnerable to the cyclone, based on the study of integrated data layers of GIS, weather & administrative boundaries



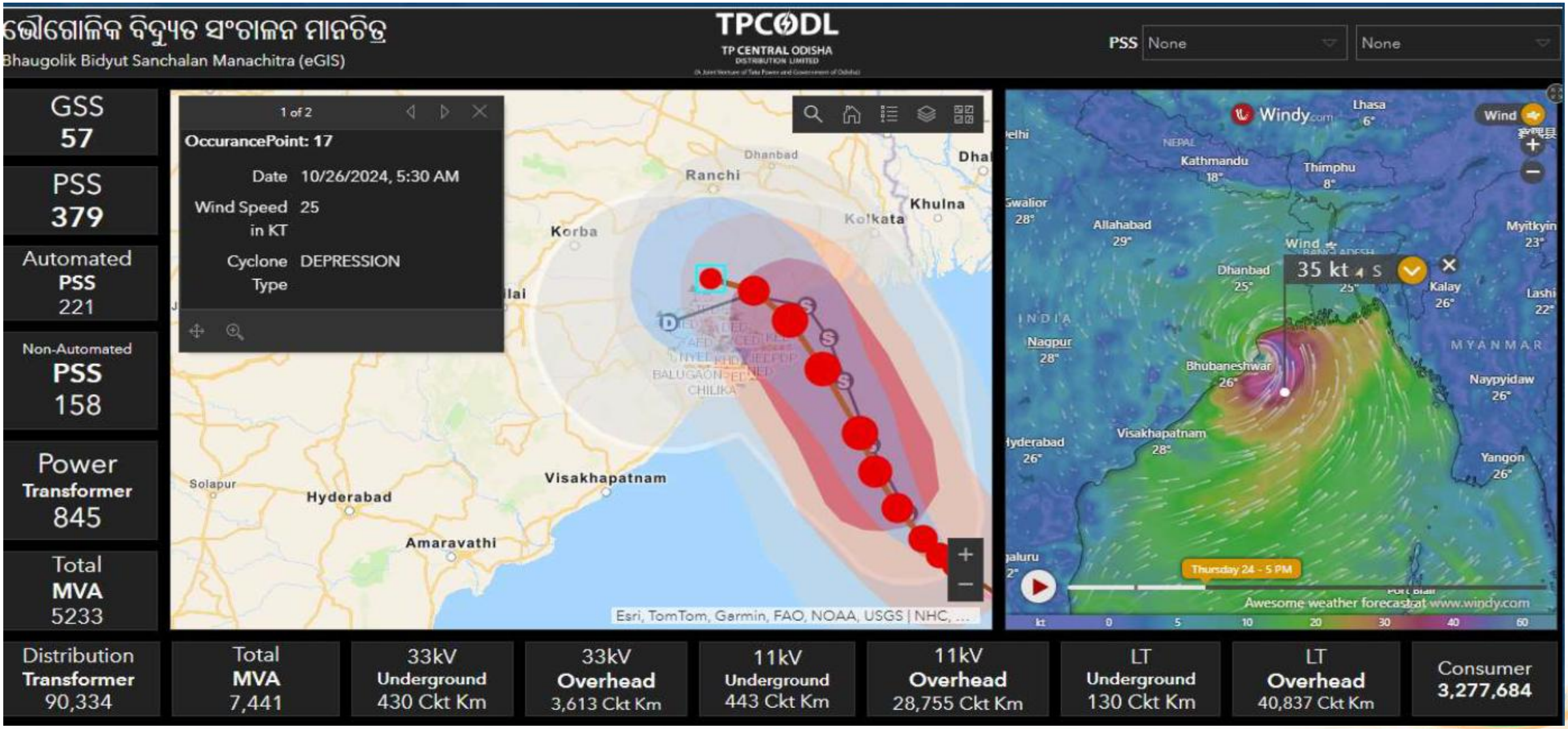
Case Study – Dana Cyclone

Location based categorization of network vulnerability assessment

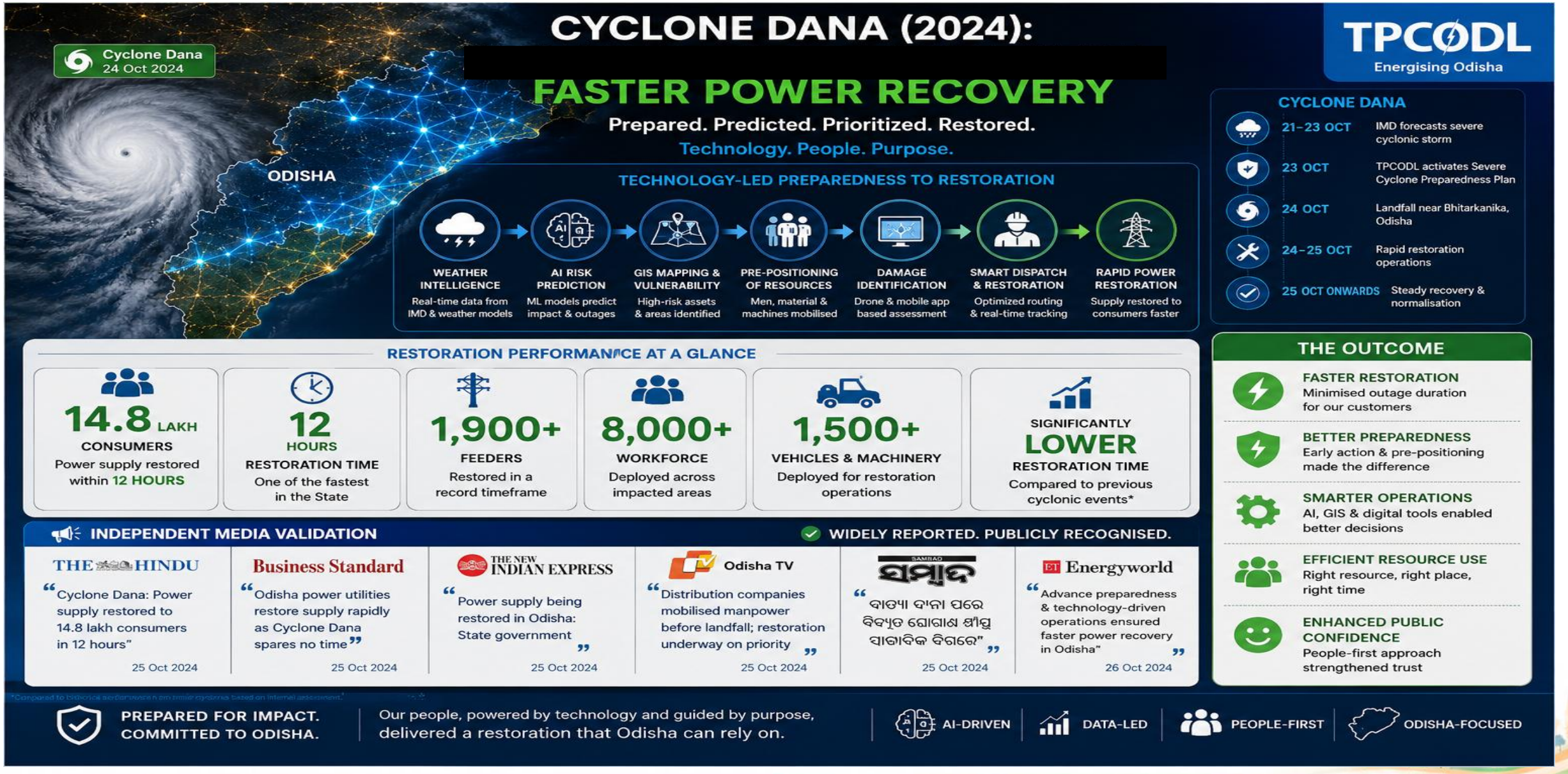


Case Study – Dana Cyclone: Tracker Aided Proactive Operations

Location based categorization of network vulnerability assessment

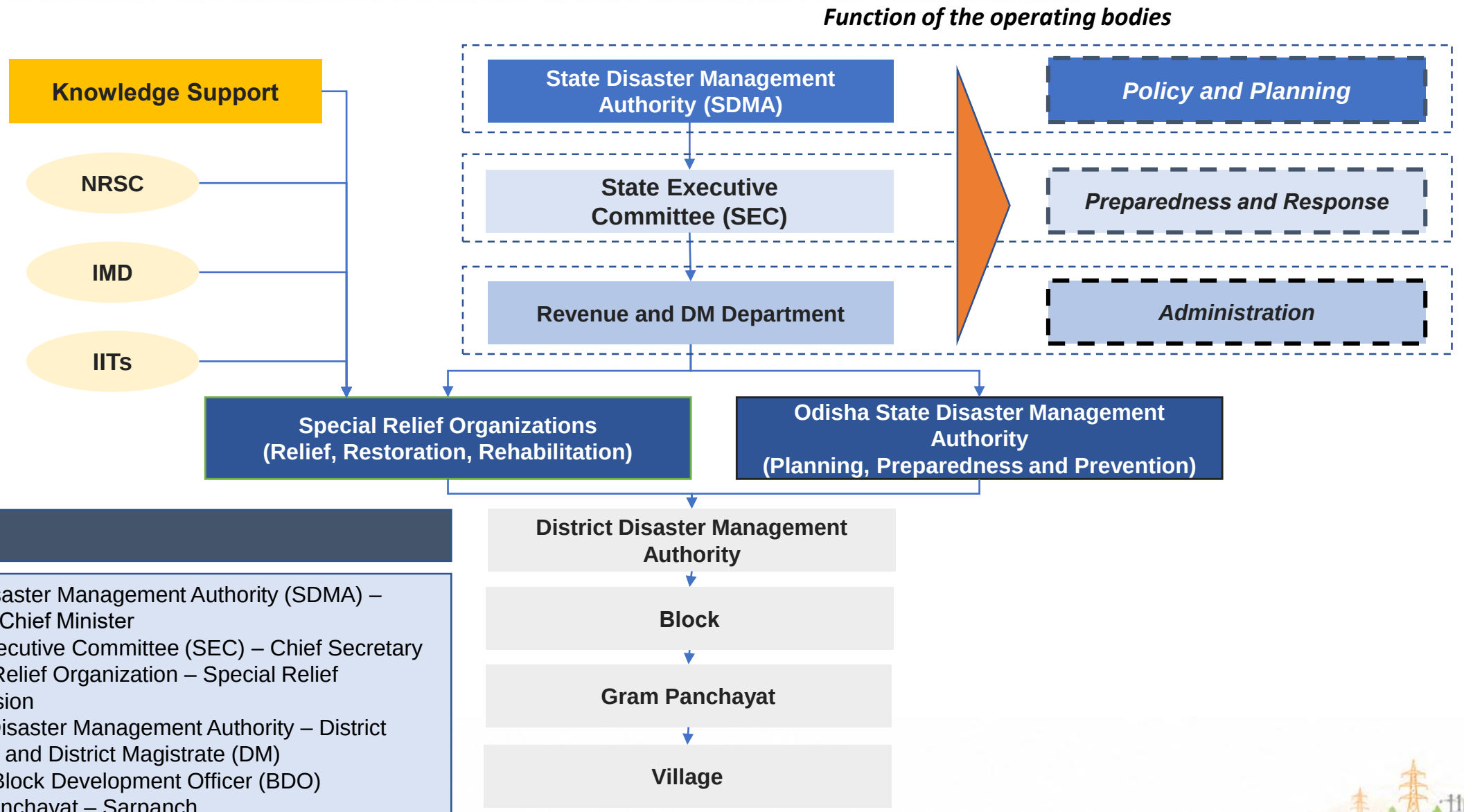


Impact of Technology Enabled Disaster Resiliency




CYCLONE
ARNAB
PREPAREDNESS
PEOPLE | NETWORK | COMMUNITIES
PLAN
PREPARE
RESPOND
RESTORE
TPCODL
TPCODL

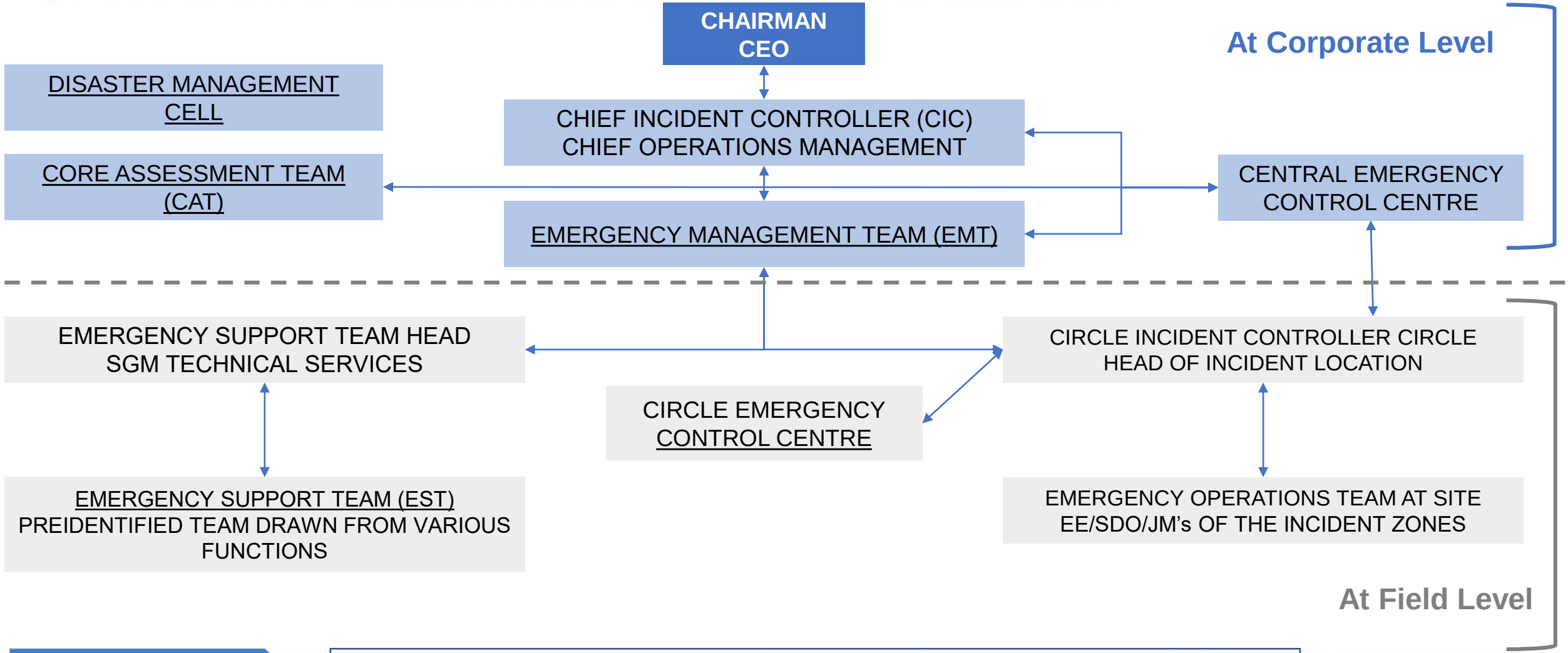
Odisha State Disaster Management



Headed by:

- State Disaster Management Authority (SDMA) – Hon'ble Chief Minister
- State Executive Committee (SEC) – Chief Secretary
- Special Relief Organization – Special Relief Commission
- District Disaster Management Authority – District Collector and District Magistrate (DM)
- Block – Block Development Officer (BDO)
- Gram Panchayat – Sarpanch

Business Continuity & Disaster Management Plan of Discom



TPCODL BCDMP framework is aligned with

- ❑ Disaster Management Act 2005
- ❑ National Policy on Disaster Management 2009
- ❑ CEA Disaster Management Plan
- ❑ UN SENDAI, SDG, COP21
- ❑ OSDMA (District DMP), Odisha

Constituents of L1, L2 & L3 Alerts

Particular	L1	L2	L3
Alert for Kalbaisakhi	Low	Moderate	Heavy
Alert for Cyclone (Km/ Hr)	62 to 87	88 to 117	Very Severe – 118 to 165 Extremely Severe – 166 to 220 Super Cyclone - > 200
Temperature trend forecast (Degree C)	> 40 for atleast 5 Hrs	> 45 for atleast 6 Hrs (Heat Wave)	> 47 (Severe Heat Wave)
Transmission network non- availability (TPCODL/ OPTCL)	Load shedding of 100 MW for 1 Hr	Load shedding of 200 MW for 1 Hr	Load shedding of 300 MW for 1 Hr
Flood Alert	Affecting 2 to 3 Divisions	Affecting 4-8 Divisions	Affecting more than 8 Divisions
No. of Consumers affected	< 50,000	< 1,00,000	> 1,00,000
No. of Interruptions (Breakdowns)	< 20	20 to 50	> 50

- **Level 1 (L1)** : Minor local incident with limited impact - local response
- **Level 2 (L2)** : Serious emergency impacting multiple locations - limited DMP activation
- **Level 3 (L3)** : Disaster disrupts complete TPCODL operations - full DMP activation

Preparedness for Exigency Situations

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Satellite Phones

Hot line connectivity & communication continuity

Availability of major items

DTs, Pole, AB Cable, Conductors

Pre & Post disaster check list

Preparedness as per BCDMP protocol

Mobile Distribution Sub Stations

For emergency & quick Power supply restoration

Emergency response teams

For critical installations such as drinking water facilities, hospitals and Collectorate

Manlifters

Tree pruning & work at height

Existing Discom Manpower

O&M manpower & additional manpower through Projects

Critical equipment

Motorized Tree pruner, de-watering pump sets



Aska lights for emergency lighting



Man lifter used for Cyclone preparedness

PDTC – Central Control Centre



State-of-Art Facility

One of its kind integrated control centre across the country

Main Control Center	**TPCODL**
Back Up Control Center	TPWODL, TPNODL & TPSODL

Real-time Monitoring through PSCC

- 24X7 Monitoring of electricity network at 33 kV level
- PSS connected through SCADA

Key Technologies Used

- SCADA and ADMS (Advanced Distribution Management System)
- GIS (Geographical Information System)
- Energy Audit
- Weather Monitoring
- Satellite Communication

Strong Data Support

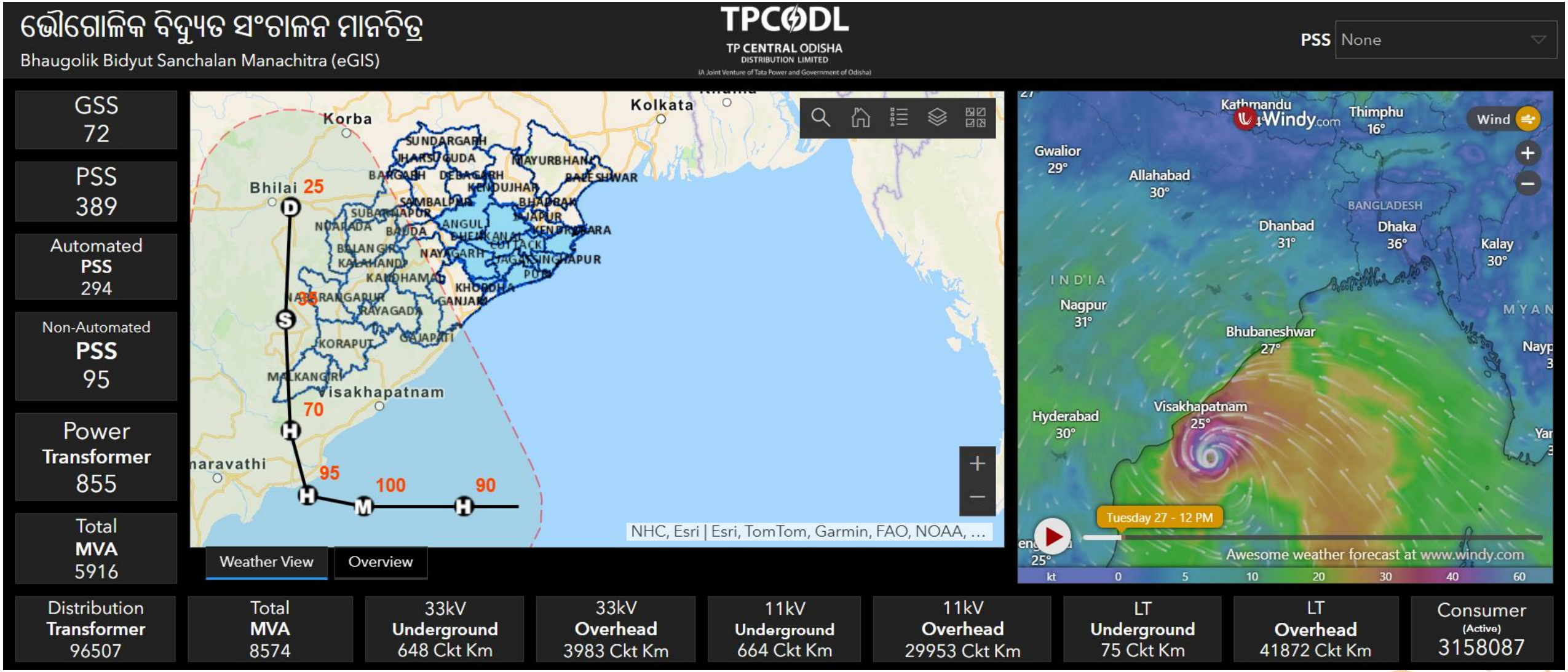
- A Data Centre with over 400 servers hosting critical systems like billing and customer apps

PDTC acts as Disaster Recovery Centre for other Odisha electricity companies, ensuring smooth operations even during emergencies.

Tracking the trajectory of Cyclone Arnab

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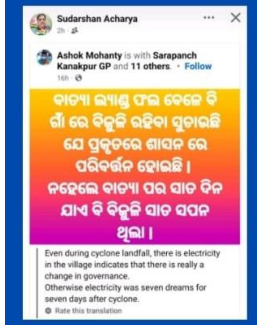
[Cyclone War Room Dashboard]

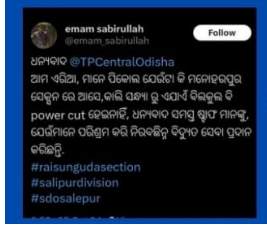
Snapshots…

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Customer Appreciation















Media Appreciation





ଶହଶହଟିର ଆବଦନ ରାଶିଛି । ଯେଉଁମାନେ ବାତ୍ୟା ଆଶ୍ରୟସ୍ଥଳୀକୁ ଆସିଛନ୍ତି ତାଙ୍କୁ ଖିପରି ଆମେ ଯନ୍ତା ଖାଦ୍ୟ ଦିଆଯିବ ତାହା ଆମର ଫୋକସ୍ ରହିଛି । ୨୨ ତାରାରେ ଗଛ ପଡି ଘାସ୍ତା ପଡି ଅବରୋଧ ହୋ...



ରାଜନଗରରେ ବାତ୍ୟା ପର ପ୍ରଭାବରେ ବ୍ୟାହତ ବିଦ୍ୟୁତ୍ ପରବରାହ ପୁନରୁଦ୍ଧାର ତାର୍ଯ୍ୟରେ TPCO...



ବିଦ୍ୟୁତ୍ ବିଭାଗର ତତ୍ପରତା

କଟକ/ପାରା, (ସମ୍ବାଦ ପ୍ରତିନିଧି): ବାତ୍ୟା 'ଦାନା' ରୁ ସଂଭାବ୍ୟ ସୁରକ୍ଷା ପାଇଁ ଆଜି ଅପରାହ୍ନ ୩ଟା ସୁଦ୍ଧା ଜିଲ୍ଲାର ୧୫ଟି ଅଞ୍ଚଳରେ ଥିବା ସମସ୍ତ ବିଦ୍ୟୁତ୍ ଯୋଗାଣ ବନ୍ଦ କରିଦିଆଯିବ ବୋଲି ଜିଲ୍ଲା ପ୍ରଶାସନ ପକ୍ଷରୁ ଘୋଷଣା କରାଯାଇଛି । ଏହି ସମୟ ସୀମା ଭିତରେ ଜିଲ୍ଲାର ୧୫ଟି ଅଞ୍ଚଳରେ ଥିବା ସମସ୍ତ ବିଦ୍ୟୁତ୍ ଯୋଗାଣ ବନ୍ଦ କରିଦିଆଯିବ ବୋଲି ଜିଲ୍ଲା ପ୍ରଶାସନ ପକ୍ଷରୁ ଘୋଷଣା କରାଯାଇଛି । ଏହି ସମୟ ସୀମା ଭିତରେ ଜିଲ୍ଲାର ୧୫ଟି ଅଞ୍ଚଳରେ ଥିବା ସମସ୍ତ ବିଦ୍ୟୁତ୍ ଯୋଗାଣ ବନ୍ଦ କରିଦିଆଯିବ ବୋଲି ଜିଲ୍ଲା ପ୍ରଶାସନ ପକ୍ଷରୁ ଘୋଷଣା କରାଯାଇଛି । ଏହି ସମୟ ସୀମା ଭିତରେ ଜିଲ୍ଲାର ୧୫ଟି ଅଞ୍ଚଳରେ ଥିବା ସମସ୍ତ ବିଦ୍ୟୁତ୍ ଯୋଗାଣ ବନ୍ଦ କରିଦିଆଯିବ ବୋଲି ଜିଲ୍ଲା ପ୍ରଶାସନ ପକ୍ଷରୁ ଘୋଷଣା କରାଯାଇଛି ।



ସମ୍ବାଦ

ଅଡ଼ଶପୁର ବିଦ୍ୟୁତ୍ କର୍ମଚାରୀ ସଜାଗ

ବଜ୍ରପାତ, ୨୪/୧୦ (ଭମିସ): ବାତ୍ୟା ମୁକାବିଲା ପାଇଁ ଅଡ଼ଶପୁର ସବ୍‌ଷ୍ଟେସନ୍ ବିଦ୍ୟୁତ୍ ବିଭାଗର କର୍ମଚାରୀ ଆଲର୍ଟ ରହିଛନ୍ତି। ଏନି ଷ୍ଟେସନ୍ ଅଧୀନରେ ଥିବା ଉପଭୋକ୍ତାଙ୍କୁ ୨୪ ଘଣ୍ଟିଆ ସେବା ଯୋଗାଇ ଦେଇଥିବା ଜଣାପଡ଼ିଛି। ବିଭାଗୀୟ ଜେଇ ଶୁଶାନ୍ତ କୁମାର ପ୍ରଧାନ ଓ ଜେଇ ବିଶ୍ୱରଞ୍ଜନ ବିଶ୍ୱାଳ, ଲାଇନ୍‌ମ୍ୟାନ୍, କର୍ମଚାରୀ ସଜାଗ ରହିଛନ୍ତି।

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APPRECIATIONS



Preparedness for the upcoming Arnab Cyclone



PDTC Geared up to handle exigencies



Restoration during night using Full-Body Harness



Restoration despite Extreme Conditions



Crossing Rivers to Restore Power

THANK YOU

For discussions/suggestions/queries email: ***dum@indiasmartgrid.org***

www.dumindia.in

Links/References (If any)

